

Employment

Graduate Research Assistant	NC State University	April 2025 – Present
- Designed cloud-based genomic platform with JBrowse 2, enabling 100+ researchers to explore 500GB datasets worldwide, deployed on Google Cloud Run. - Collaborated on ML architecture design, applying hyperparameter tuning and time-series analysis, improving model accuracy by 12%.		
Product Engineer Intern	Isomer AI	May 2025 – Present
- Provisioned cloud infra with Terraform; owned full backend: event pipelines, Temporal.io workflows, PostgreSQL, Redis, Docker, Auth0, Sentry; shipped 10+ features via CI/CD across 7 deployments in TypeScript. - Engineered multi-model LLM tooling (Claude, GPT-4o, Gemini) via MCP with provider-agnostic interfaces; built document processing (PDF gen, email parsing, S3) and Playwright test coverage.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Automated deployments on AWS and GCP, integrated Grafana monitoring dashboards, and led model red-teaming to strengthen system reliability. - Built 5+ AI-driven workflows including health claim automation using OCR, NLP, and ML; improved claim processing accuracy by 25%.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for a data-driven system automating support ticket analysis, reducing processing time by 15%. - Built distributed pipelines with Hadoop + Spark for batch processing and Kafka streaming for real-time ingestion, cutting latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

PulseRank	Real-Time Leaderboard	2026
- Shipped OpenTelemetry + Prometheus + Grafana observability stack on all hot paths; load-tested with k6 and deployed via GitHub Actions CI/CD. - Built real-time leaderboard with MySQL outbox, MongoDB TTL projections, and Redis sorted sets for sub-ms reads.		
Demand Paging – Xinu OS	Systems Programming	Oct 2024 – Dec 2024
- Engineered 8-store virtual memory system in C for Xinu OS with Second-Chance page replacement; reduced page faults by 25%. - Added 7+ kernel system calls, increasing memory performance by 15% using inverted page tables and a custom fault handler.		
VectorLift	Semantic Search Engine	2026
- Evaluated BM25 (Elasticsearch), dense bi-encoder with in-batch negatives, and cross-encoder re-ranker on MS MARCO. - Scored NDCG@10, MRR@10, MAP; served via FastAPI with per-stage latency tracking; Kafka-driven corpus indexing into Qdrant.		
LLM Eval Harness	Independent Project	2025
- Built vendor-neutral LLM benchmark comparing Claude, GPT-4o, and Gemini across 2,000 examples and 3 task types. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		

Languages and Technologies

- Languages:** Python, TypeScript, Java, JavaScript, C, Go, Bash, SQL
- Infra & DevOps:** Terraform, Docker, Kubernetes, AWS, GCP, CI/CD pipelines, Linux, GitHub Actions
- Observability:** OpenTelemetry, Prometheus, Grafana, Sentry, Distributed Tracing, Alerting, Kafka
- Backend/Stack:** Node.js, NestJS, Temporal.io, FastAPI, PostgreSQL, Redis, Elasticsearch, Spark, Hadoop

Additional Experience and Awards

- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.